

Quiz 1 Solution

Q1.

```
1 class Car {
2     private:
3         string model;
4         int price;
5
6     public:
7         Car() {}
8         Car(string m, int p) : model(m), price(p) {}
9
10        Car(const Car& other) {
11            if (this == &other) return;
12
13            this->model = other.model;
14            this->price = other.price;
15        }
16
17        Car operator=(const Car& other) {
18            if (this == &other) return *this;
19
20            this->model = other.model;
21            this->price = other.price;
22            return *this;
23        }
24    };
25
26    int main() {
27
28        Car obj1("Hyundai", 200);
29        Car obj2 = obj1;
30
31    }
32
```

Following is the code defining a Car Api, which function is invoked on line 29?

- a) Car() and Car(const Car& other)
- b) Car() and Car operator=(const Car& other)
- c) Car(const Car& other)
- d) Car operator=(const Car& other)

Correct Answer: option c

Explanation: obj1 is already declared and instantiated at line 28, therefore at line 29 when obj2 is declared, it looks for a valid constructor for it to be initiated, it is provided another Car object,

which reflects with its constructor `Car(const Car& other)` and therefore the copy constructor is called.

Q2.

```
class Students {  
private:  
    string* students;  
public:  
    Students() {  
        students = new string[100];  
    }  
};
```

The following class declares dynamic memory for 100 students, your task is to write a destructor. Which one of the following will release all memory held by `*students`.

- a) `delete[] students;`
- b) `for(int i=0; i<100;i++){ delete students[i]; }`
- c) `delete students;`
- d) both a) and b).

Correct Answer: option a

Explanation: The constructor dynamically allocates an **array** of 100 string objects using `new string[100];`.

To properly release this memory, you **must** use `delete[] students;` in the destructor.

Option (b) (`for(int i=0; i<100; i++) { delete students[i]; }`) is incorrect because `students[i]` is not a pointer, but an actual string object. This would result in undefined behavior.

Q3. Which of the following is NOT possible inside a const member function of a class?

- a) modify an attribute of the calling object
- b) modify an attribute of an object of the same class passed as an argument
- c) modify an attribute of an object of a different class passed as an argument
- d) modify a local variable of the function

Correct Answer: option a

Explanation:

Declaring a member function as const makes the calling object read only, access to change class variables is taken away from user for that object, if user still tries to access and change the variable, the program raises a compilation error.

Q4. The following is an implementation of an Object class

```
class Object{
private:
    inline static int count = 0;
public:
    Object() { count++; }

    int getCount() { return count; }
};

int main()
{
    Object o1;
    Object o2;
    cout << o1.getCount();
    Object o3;
}
```

What will be the output of the program?

- a) 0
- b) 2
- c) 3
- d) 1

Correct Answer: option b

Explanation:

Static declaration of data members share an instance across all instances of an object, any change in one is represented in all, the first two object increment the count to +2, the print statement prints 2, although count is later updated to 3 since another object is instantiated but we don't get to see that since the cout statement is before the updation.

Q5. If a Class has no access specifier defined for its members, then the members are automatically assumed to be at which access level?

```
class Date{  
    /* no access specifier */  
    int day;  
    int mon;  
    int year;  
  
    /*member functions*/  
};
```

- a) protected
- b) public
- c) private
- d) none

Correct Answer: option c

Explanation:

In C++ classes if access specifier is not defined then the data members automatically assume private access specifier.

Q6. Which of the following best describes encapsulation in C++?

- A) Hiding the implementation details of a class and exposing only necessary functionalities
- B) The ability of different objects to respond to the same message in different ways
- C) Creating a copy of an existing object
- D) Defining multiple functions with the same name but different parameters

Answer: A

Q7. What will be the output of the following code?

```
class Sample {  
    private:  
        int data;  
    public:  
        void setData(int d) { data = d; }  
        int getData() { return data; }  
};  
  
int main() {  
    Sample obj;  
    obj.setData(10);  
    cout << obj.data;  
    return 0;  
}
```

- A) 10
- B) 0
- C) Compilation Error
- D) Garbage Value

Answer: C

Q8. Why are getters and setters used in classes?

- A) To directly access class members
- B) To restrict modification of private variables to within the class
- C) To reduce memory usage
- D) To increase the speed of program execution

Answer: B

Q9. Which of the following statements about constructors in C++ is FALSE?

- A) A constructor is automatically called when an object is created
- B) A constructor can have a return type
- C) Constructors can be overloaded
- D) A class can have multiple constructors with different parameters

Answer: B

Q10. When is the copy assignment operator called in C++?

- A) When a new object is assigned another object of the same class
- B) When another object of the same class is passed as an argument to a new object
- C) When an object of the same class is assigned to another object
- D) All of the above

Answer: C